

DAY 09 – LINUX BASICS

Name:- Rishabh Raj

1. What is Linux?

Linux is an open-source operating system kernel. It is used as the core of many operating systems and is popular on servers, desktops, cloud systems, embedded devices, and cybersecurity platforms.

2. What is a Linux distribution?

A Linux distribution (distro) is a complete operating system built around the Linux kernel. It usually includes system utilities, package management tools, libraries, a desktop environment or command-line tools, and applications.

3. Name four Linux distributions.

Four examples are Ubuntu, Debian, Fedora, and Kali Linux.

4. What is Kali Linux?

Kali Linux is a Debian-based Linux distribution designed mainly for cybersecurity work, including penetration testing, digital forensics, security research, and vulnerability assessment. It includes many security tools.

5. What is a terminal?

A terminal is a text-based interface through which a user interacts with the operating system by entering commands. In Linux, it is commonly used to manage files, run programs, configure systems, and perform administrative or security tasks.

6. What does the whoami command do?

The whoami command displays the username of the currently logged-in user. It is useful for confirming which account is being used in the current terminal session.

7. What does hostname display?

The hostname command displays the name assigned to the computer or system on the network. This name helps identify the machine.

8. What does pwd mean?

pwd stands for 'Print Working Directory'. It displays the full path of the directory in which the terminal is currently working.

9. What is the purpose of ls?

The ls command lists files and directories in the current directory or in a specified directory. It is one of the basic commands used for navigating and inspecting the Linux file system.

10. Explain the difference between ls, ls -l and ls -a.

ls shows the normal visible files and directories. ls -l shows a detailed, long-format listing, including permissions, ownership, size, and modification time. ls -a shows all entries, including hidden files and directories whose names normally begin with a dot (.).

11. What does cd do?

cd stands for 'change directory'. It changes the terminal's current working directory to another directory.

12. What does cd .. do?

cd .. moves the current working directory one level up to the parent directory.

13. What does cd ~ do?

`cd ~` takes the user to their home directory. The tilde (~) is a shortcut representing the current user's home directory.

14. What does `cd /` do?

`cd /` takes the user to the root directory, which is the top-level directory of the Linux file system.

15. What is the Linux root directory?

The Linux root directory is represented by `/`. It is the top-level directory of the file-system hierarchy. Important directories such as `/home`, `/etc`, `/var`, `/usr`, and `/bin` are located under it.

16. What is the difference between the root directory and the root user?

The root directory `/` is the top level of the file system. The root user is a special administrative account with very high privileges. They are different concepts: one is a location in the file system and the other is a user account.

17. What is an absolute path?

An absolute path gives the complete location of a file or directory starting from the root directory `/`. For example, `/home/student/Documents/file.txt` is an absolute path.

18. What is a relative path?

A relative path describes a file or directory location from the current working directory. It does not start from `/`. For example, `Documents/file.txt` is a relative path if `Documents` exists inside the current directory.

19. How are hidden Linux files commonly identified?

Hidden Linux files are commonly identified by a dot (.) at the beginning of their name, such as `.bashrc`. They are not normally shown by a basic `ls` command; `ls -a` can display them.

20. Why are Linux skills important for cybersecurity?

Linux skills are important in cybersecurity because many servers, security appliances, cloud systems, and security tools use Linux. Security professionals frequently use the terminal for reconnaissance, log analysis, scripting, system administration, network investigation, penetration testing, and digital forensics.

PRACTICAL – COMMANDS & SCREENSHOTS

Run the following commands on your own Linux system/VM (for example, Kali Linux or Ubuntu) and take a screenshot of the terminal after running them. Paste your own screenshots in the spaces provided.

whoami – Displays the current username.

whoami

```
Last login: Thu Sep 17 10:40:15 on console
rishabhraj@Rishabhs-MacBook-Air ~ % whoami
rishabhraj
rishabhraj@Rishabhs-MacBook-Air ~ %
```

hostname – Displays the computer/system hostname.

hostname

```
rishabhraj@Rishabhs-MacBook-Air ~ % hostname
Rishabhs-MacBook-Air.local
rishabhraj@Rishabhs-MacBook-Air ~ %
```

pwd – Displays the current working directory.

pwd

```
rishabhraj@Rishabhs-MacBook-Air ~ % pwd
/Users/rishabhraj
rishabhraj@Rishabhs-MacBook-Air ~ %
```

ls – Lists visible files and directories.

ls

```
rishabhraj@Rishabhs-MacBook-Air ~ % ls
All Images           Downloads            Pictures
All Project here     E_Image             Postman
APCSIP-2026          Important Image.sparsebundle  Public
Applications         Library             RISHABH
College              Movies              satyam
Desktop              Music               shell.php
Documents            MYFiles_R           VirtualBox VMs
rishabhraj@Rishabhs-MacBook-Air ~ %
```

ls -l – Shows a detailed listing.

ls -l

```
rishabhraj@Rishabhs-MacBook-Air ~ % ls -l
total 8
drwxr-xr-x  5 rishabhraj  staff   160  4 Jul 12:20 All Images
drwxr-xr-x 13 rishabhraj  staff   416 11 Sep 14:05 All Project here
drwxr-xr-x 42 rishabhraj  staff  1344  4 Jul 12:11 APCSIP-2026
drwx-----@ 7 rishabhraj  staff   224 26 Mar 22:48 Applications
drwxr-xr-x  7 rishabhraj  staff   224 27 Jul 00:49 College
drwx-----@ 29 rishabhraj  staff   928 17 Sep 17:06 Desktop
drwx-----@  6 rishabhraj  staff   192  2 Sep 19:23 Documents
drwx-----+ 261 rishabhraj  staff  8352 17 Sep 11:33 Downloads
drwxr-xr-x 319 rishabhraj  staff 10208  4 Jul 12:07 E_Image
drwxr-xr-x@  8 rishabhraj  staff   256  1 May 00:34 Important Image.sparsebundle
drwx-----@ 96 rishabhraj  staff  3072 13 Mar 2026 Library
drwx----- 15 rishabhraj  staff   480 15 Sep 23:44 Movies
drwx-----+  8 rishabhraj  staff   256  4 Apr 00:16 Music
drwxr-xr-x  5 rishabhraj  staff   160  8 Sep 14:11 MYFiles_R
drwx-----+  7 rishabhraj  staff   224  4 Mar 2026 Pictures
drwxr-xr-x@  4 rishabhraj  staff   128 18 Feb 2026 Postman
drwxr-xr-x+  5 rishabhraj  staff   160 18 Feb 2026 Public
```

ls -a – Shows visible and hidden files/directories.

ls -a

```
rishabhraj@Rishabhs-MacBook-Air ~ % ls -a
.          .osint_tool          Documents
..         .putty              Downloads
.android   .python_history         E_Image
.BurpSuite .ssh                        Important Image.sparsebundle
.cache     .Trash                     Library
.CFUserTextEncoding .vscode                    Movies
.claude    .vscode-shared         Music
.claude.json .zprofile                  MYFiles_R
.codex     .zsh_history              Pictures
.config    .zsh_sessions             Postman
.copilot   All Images          Public
.DS_Store  All Project here       RISHABH
.idlrc     APCSIP-2026              satyam
.junie     Applications       shell.php
.local     College           VirtualBox VMs
```

cd / – Moves to the root directory.

cd /

```
rishabhraj@Rishabhs-MacBook-Air ~ % cd /
rishabhraj@Rishabhs-MacBook-Air / %
```

cd .. – Moves one level up to the parent directory.

cd ..

```
rishabhraj@Rishabhs-MacBook-Air / % cd ..
rishabhraj@Rishabhs-MacBook-Air / %
```

cd ~ – Moves to the user's home directory.

cd ~

```
rishabhraj@Rishabhs-MacBook-Air / % cd ~
rishabhraj@Rishabhs-MacBook-Air ~ %
```

